



CLEVELAND-CLIFFS: HEATING STEEL WITH STATE-OF-THE-ART, ENERGY-EFFICIENT FURNACES

SECTOR

Industrial

LOCATION

Burns Harbor

FINANCIAL OVERVIEW

\$160 million

BACKGROUND

The Cleveland-Cliffs' Burns Harbor, Indiana, steel plant began operation in 1966 as a four-million-ton hot strip mill with three pusher-style reheat furnaces. The furnaces heat steel slabs for rolling from ambient temperature to 2,300 degrees Fahrenheit. After five decades in operation, the Burns Harbor hot strip mill pusher-style furnaces needed to be replaced, opening up an opportunity to reduce energy usage and emissions as well as to improve overall mill productivity.

The existing furnaces were replaced with walking beam reheat furnaces, resulting in over 2,100,000 MMBtu of natural gas savings per year and approximately \$8.4 million in fuel savings. Additional benefits include improved product quality and reduced maintenance, which are expected to result in a total annual benefit of \$18.8 million from the project.

SOLUTIONS

The goal of the Burns Harbor project was to install two new, state-of-the-art walking beam reheat furnaces to replace the original three pusher furnaces, with minimal disruption to production. Cleveland-Cliffs decided to replace the three pusher furnaces with two long walking beam furnaces. The new furnace system eliminates the refractory hearth process and now each slab is “walked” through instead of pushed, eliminating the bottom side defects and laminations commonly experienced with pusher furnaces. Due to the high capital cost of the project—\$160 million—multiple objectives were needed for this project to be approved. These included:

1. Improve energy efficiency and reduce greenhouse gas emission generation at the Burns Harbor Mill by reducing fuel consumption by 30%
2. Improve the bottom side slab/strip quality by eliminating the skid and hearth mechanical friction and thermal impact with the slabs
3. Achieve an annual production of 4.2 million tons/year with a peaking capacity of 5.0 million tons/year when required
4. Expand surface critical product offerings for automotive and non-automotive business
5. Install and operate state-of-the-art furnace safety controls
6. Automate the slab charging and de-piling functions

Improving product quality and productivity was Cleveland-Cliff's primary reason for replacing the pusher furnaces. The business case was developed over a decade and the installation took place over four years. Once site preparation was complete with underground and overhead utility relocations, furnaces were ordered, and the mechanical and electrical installations were completed. An international equipment provider supplied two new top and bottom-fired walking beam furnaces. This included the design, supply, erection supervision, and commissioning of the new furnaces as well as comprehensive training of the operating and maintenance personnel. Each of the two furnaces will heat slabs at a rate of 500 tons per hour. The combustion systems include low-NOx burners that will reduce NOx emissions compared to the previous burners. The furnaces will reduce overall fuel consumption because of their long, unfired preheat zones, individual burner control, and improved insulation, which includes efficient skid insulation. Local industrial construction companies completed the mechanical, structural, and piping installation of the furnaces. The two new furnaces weigh a combined 2,670 tons, including skids and posts.



Figure 1. The first slab was extracted from the #5 Walking Beam Furnace on March 23, 2021.

Below are the estimated annual benefits from the installation of the walking beam furnaces. These numbers assume the hot strip mill processes 4.2 million tons of steel annually.

Annual Benefit (Recurring)	Financial Benefits
Energy (natural gas + power)	\$7,500,000
Quality (reprocessing, process yield, and claims)	\$4,040,000
Productivity (Increased delay free tons per hour reduces required run time and overtime costs)	\$1,900,000
Labor, Repair & Maintenance Parts/Maintenance/Contractors/Roll stock consumption	\$5,380,000
Total Annual Benefits	\$18,820,000

Cleveland-Cliffs measured the following metrics to evaluate the project’s success:

Key Success Goals	2017 Average	4Q 2021	Details
Natural Gas Consumption	1.7 MMBtu/ton	1.369 MMBtu/ton	The first full quarter after commissioning showed a 19.5% improvement in gas consumption; however, the new furnace system has demonstrated it is capable of heating slabs at less than 1.2 MMBtu/ton.
Mill-Related Rejects	1.2%	0.67%	35.3% reduction of in-house rejects—surpassing the post WBF goal.
Holes and Lamination Rejects	0.460%	0.083%	Holes and lamination rejects are directly related to pusher furnaces. An 82% reduction in quality issues is a reduction in energy consumption due to wasted energy that was used in rolling a rejected product.





Figure 2. Waste gas ducts, air, and natural gas piping.



Figure 3. Refractory installation in the bottom of the furnace.

OTHER BENEFITS

Cleveland-Cliffs Burns Harbor further measured the success of the walking beam furnaces by tracking reductions in electric power generation, greenhouse gas emissions, product quality issues, and repair and maintenance costs. The anticipated natural gas reduction of 2,100,000 MMBtu will result in a reduction of over 109,000 metric tons of CO₂e per year. The overall productivity of the mill has improved significantly, with record tonnage rolling in an eight-hour turn and 6,659 tons rolled at 823.3 tons per hour, breaking the previous record held for 46 years.

IMAGE GALLERY



WALKING BEAM FURNACE